



This Assignment will not be considered for Course participation Marks after 24th march 2024

Instructions for Extended Match Questions

Use this slide to understand how to answer an EMQ

Instructions:

1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. [Please click here to view a sample template file for answering.](#)
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

List of Options:

1. $\dim(V) = 1$
2. $\dim(V) = 2$
3. $\dim(V) = 3$
4. V is not a vector space
5. V a vector subspace of $\mathbb{R}^3$
6. $\{(1, 0, 1), (0, 1, 1)\}$ is a basis of V .
7. $\{(1, 0, 1), (0, 1, -1)\}$ is a basis of V .
8. $\{(1, 1, 0), (1, 0, 1)\}$ is a basis of V .
9. $\{(-1, 1, 0), (1, 0, 1)\}$ is a basis of V .

List of Questions:

- Q1.** Consider the set $V = \{(a, b, c) \mid a + b = c \text{ and } a, b, c \in \mathbb{R}\}$ with usual coordinate-wise addition and scalar multiplication. Which of the above options is(are) true for V ?
- Q2.** Consider the set $V = \{(a, b, c) \mid a = b + c \text{ and } a, b, c \in \mathbb{R}\}$ with usual coordinate-wise addition and scalar multiplication. Which of the above options is(are) true for V ?
- Q3.** Consider the set $V = \{(a, b, c) \mid a + b = 1 - c \text{ and } a, b, c \in \mathbb{R}\}$ with usual coordinate-wise addition and scalar multiplication. Which of the above options is(are) true for V ?